

PRECISION FLOW CONTROL
& GREENHOUSE AUTOMATION

Precision Flow Control & Greenhouse Automation is a practical engineering guide designed for smart farm operators, greenhouse technicians, and vocational students (including Gärtner Ausbildung candidates). This manual bridges the gap between agricultural physics and daily greenhouse operations, offering step-by-step instructions and verified control designs.

KEY TECHNICAL TOPICS COVERED:

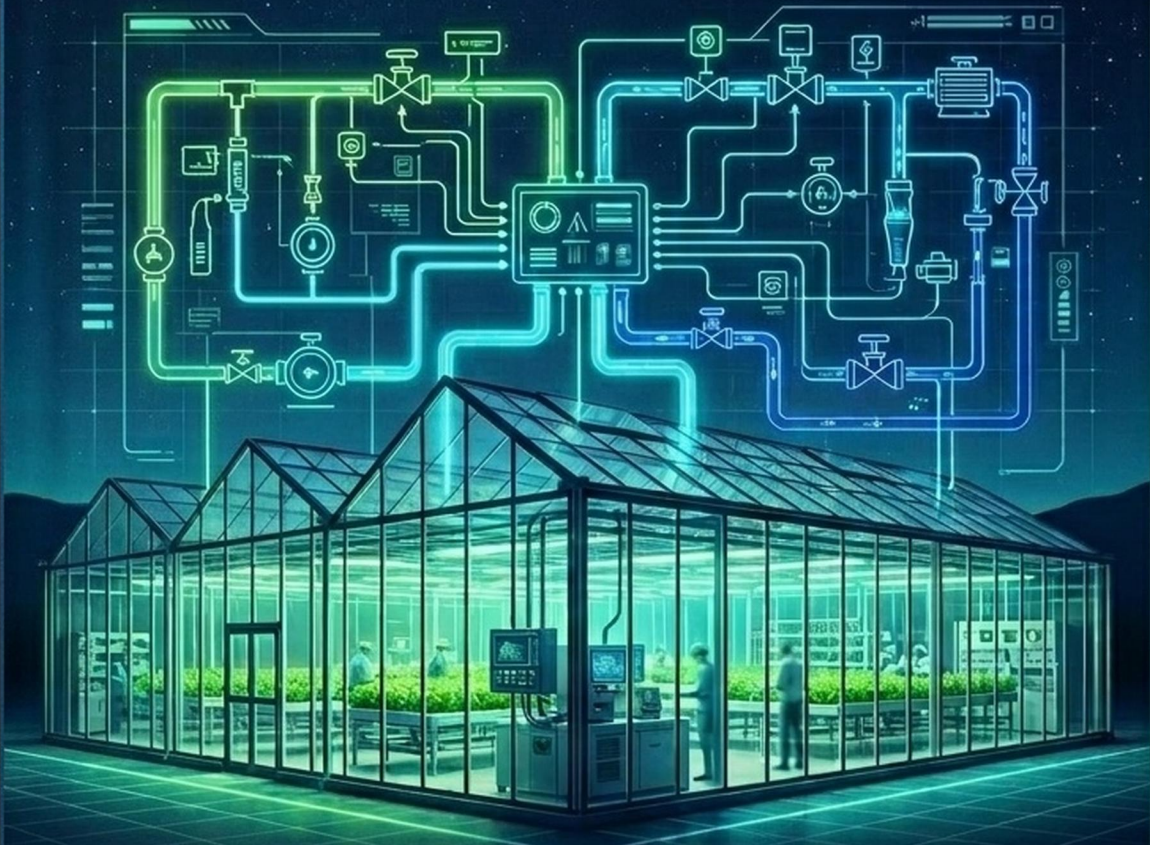
- Venlo Glass Greenhouse Structural Load & Light Transmittance Modeling
- Buoyancy-Driven Convection Dynamics & Destratification Calculations
- Proportional Mixing Valves & High-Temperature Boiler Loop Control
- Structured Text (ST) PLC Code for PIDs with Anti-Windup Clamping
- EC/pH Sensor Calibration Protocols & Ultrasonic Flow Meters
- Nitric/Phosphoric Acid Dosing Safety Interlocks & Dilution Rules
- Closed-Loop Water Recirculation, UV-C Disinfection, & LSI Scale Mitigation

About the Author

Inwoo Hwang is a horticultural crop science researcher and engineering consultant with field experience in high-temperature fluid systems (Dawoovac) and international logistics (DBeep, Netherlands). He specializes in AgTech system automation and greenhouse microclimate design.

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PRACTICAL MANUAL FOR SMART FARM
MAINTENANCE AND OPERATIONS



By Inwoo Hwang
TECHNICAL ENGINEERING GUIDE